

(Transaction) Logs

What is a log?

Record of actions

- Written to stable storage
- Can use to restore consistent state

- E.g., (TID, location, old_value, new_value)

Why need a log?

Why not just do all actions at once at end of transaction?

- Actions take time – failure could interrupt
- Transaction could use lots of space
- Transaction could use lots of time

Key ideas

- Redo committed changes from a valid state
- Undo uncommitted changes from current state
- Checkpoint: a valid state
  - Big and expensive
  - Can ignore or throw away prior log entries

- Want efficiency, but writing to disk is slow.